

Business And Society

Is AI a Boom or a Bubble?

by Paulo Carvão

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Summary. Nvidia's \$100 billion commitment to OpenAI—and OpenAI's reciprocal chip purchases—signal an AI arms race accelerating at breakneck speed. How can leaders make sense of this market? An examination of recent history (particularly

the parallels between this moment and the recent dot-com bubble) offers some clues. Speculative capital and circular financing can distort timing and expectations, particularly where AI adoption remains uneven. Currently, consumer enthusiasm outpaces enterprise integration, raising the risk of overcapacity. Durable advantage will come from disciplined investment, embedding AI into workflows, and shaping governance to balance innovation with resilience.

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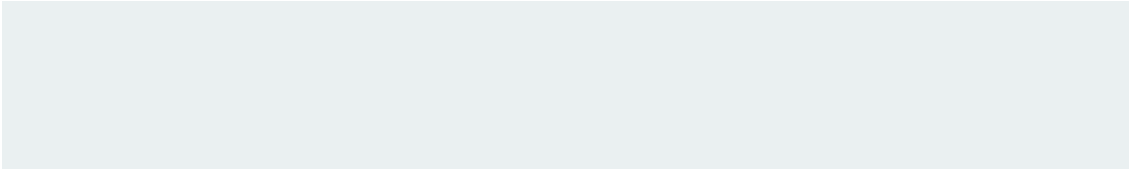
In late September, Nvidia [announced](#) plans to invest up to \$100 billion in OpenAI to fund a new generation of data centers, while OpenAI [pledged](#) to purchase millions of Nvidia chips for those facilities. Days later, OpenAI [struck](#) a similar multibillion-dollar arrangement with AMD. Celebrated by investors, these deals also raised eyebrows. To some observers, they looked eerily like the circular financing arrangements of the late 1990s, when vendors and clients reinforced each other's valuations without generating real value. [Bloomberg](#) aptly described the pattern as an “increasingly complex and interconnected web of business transactions” fueling a trillion-dollar AI boom.

The rapid buildout of AI is reshaping executive decision-making and compressing timelines. Not having a clear perspective on AI is no longer an option. Capital is moving fast, infrastructure is



being committed at unprecedented scale, and competitive pressures are intensifying.





In this new environment, speed and intent matter, and the companies that will thrive are those that commit resources where returns are durable, identify which trends to embrace, and avoid bets that won't hold. This article will trace the forces driving the boom, discuss the risks of a bubble, and outline the practical choices leaders face as they weigh opportunity against exposure.

Innovation Meets Speculation

The core challenge facing leaders today is how to balance AI's long-term promise with the short-term pressure to deliver returns. The AI economy is being propelled by two races: a geopolitical contest for supremacy and a financial race to generate the returns necessary to justify staggering levels of investment. When these two forces converge, they amplify one another, heightening the risk of overshoot.

The parallels with earlier bubbles are striking. During the dot-com era, capital flooded into telecom infrastructure, betting on exponential internet demand. When that demand lagged, bankruptcies cascaded across the sector. The problem was not



that the internet lacked potential, but that the timing of capital deployment outpaced adoption.

Today, we see similar signals. Market concentration reached alarming levels: the “Magnificent Seven” technology firms account for more than one-third of the S&P 500, a level twice as high as that of the top tech companies during the 2000 bubble. Valuations for AI companies are based on aggressive projections rather than current earnings. Complex deal structures, such as OpenAI’s financing through vendor agreements, resemble the circularity of past bubbles. Meanwhile, vast sums are being committed to data centers, chip supply chains, and foundational models even as enterprise adoption proceeds more cautiously than consumer enthusiasm.

Adoption has split. Consumers embraced ChatGPT at record speed, reaching 100 million users faster than any prior app. Enterprises remain hesitant. MIT found productivity gains elusive at scale and advised focusing on high-value uses, embedding AI in workflows, and scaling gradually. According to Stanford’s AI Index, business adoption rose to 78% in 2024 from 55% the previous year. Yet companies remain tentative, citing privacy, reliability, compliance, security, and financial risk as key barriers.



The hype around consumer use has yet to translate into broad enterprise change. The infrastructure build assumes that adoption will follow, yet enterprises rarely commit without a line of sight to financial returns. If demand lags, today's massive buildout of chips and data centers could create overcapacity. This dynamic is further exacerbated by industrial policy and national security priorities that prioritize investment ahead of consumption, contributing to growth but also amplifying the risk of a bubble.

Industrial Policy, Capital Flows, and the Geopolitics of AI

The hand of government has shaped America's technology cycles before, often by creating incentives, making targeted investments, and then stepping aside to let industry scale. In the 1990s, Washington drove the internet's expansion by privatizing network infrastructure, passing the 1996 Telecommunications Act to open markets, and enacting Section 230 to limit platform liability. Those choices created the conditions for growth, investment, and innovation. Today, policymakers are applying a similar playbook to artificial intelligence. Both the Trump and Biden administrations have emphasized industrial policy, framing AI not only as an economic opportunity but also as a national security imperative. The implicit message for businesses is that speed is more important than caution.



China's state-led model channels capital into AI to build domestic champions and cut reliance on U.S. technology. Europe has focused on risk management, but fears of overregulation prompted CEOs to warn of lost competitiveness. In response, Brussels launched the AI Continent Action Plan and a €1 billion Apply AI initiative to accelerate adoption, lower compliance costs, and turn Europe into a global AI leader.

At the same time, venture funding for AI startups exceeds what was seen in previous cycles of crypto and mobility tech, with sovereign wealth funds adding another wave of investment. These actors are deliberately investing ahead of adoption, betting that demand will eventually materialize. If it doesn't, oversupply could leave behind stranded assets, as it did with unused telecom fiber after the dot-com crash.

Enterprises, however, face a different decision. For them, the challenge is not whether to bankroll the next infrastructure wave, but how to adopt AI in ways that strengthen their own business. Companies that survived the dot-com collapse, such as Amazon, which focused relentlessly on building the infrastructure for e-commerce, did so by tying technology investment to the business, not to speculative momentum.

Understanding the Constraints



Capital and policy are speeding the AI race, but constraints limit its pace. Land, labor, and energy place hard boundaries on how quickly adoption can scale. These limits create scarcity: early movers secure capacity and edge while laggards fall behind.

To understand how these constraints affect AI's trajectory, it helps to revisit the idea of the AI triad: compute, data, and talent. These elements defined the last decade of AI development. A new triad has emerged: land, labor, and energy. Data centers need large sites with access to power and water, and communities are resisting their footprint. Labor shortages extend beyond engineers to skilled trades such as electricians, construction workers, and technicians. Energy is the toughest constraint: Goldman Sachs projects U.S. data center consumption will double by 2030, threatening the economics of scaling without sustainable solutions.

AI may be a general-purpose technology, but its infrastructure is not infinitely scalable. For companies, this means factoring the new triad directly into strategy. Securing long-term energy contracts, investing in workforce partnerships with trade schools, and working with local communities on land use can all reduce exposure to bottlenecks. Firms that anticipate these pressures and lock in capacity early will be better positioned to capture durable returns than those that assume unlimited scalability.



Is This a Bubble?

To be sure, not everyone sees a bubble. Nvidia CEO Jensen Huang argues that AI demand is structural rather than speculative, comparing today's buildout to the "beginning of a new industrial revolution."

In an interview with Ben Thompson at Stratechery, Sam Altman dismissed concerns that today's AI investments are unsustainable, framing OpenAI's strategy as moving beyond chatbots to agentic AI and a platform linking infrastructure and applications. "It's brutally difficult to have enough infrastructure in place to serve the demand," he noted, underscoring the tension between front-loaded capital and delayed returns.

In his 2024 book, *Co-Intelligence: Living and Working with AI*, Ethan Mollick discusses how AI produces practical, stepwise improvements in workplaces and classrooms. By augmenting human ability, producing small, cumulative gains, AI gradually builds toward a broader impact.

The divergence in views reflects the central uncertainty since bubbles hinge less on a technology's long-term impact and more on whether adoption and capital cycles align. The internet was revolutionary, but that did not prevent the dot-com collapse. AI



may well reshape global industries, but its financial foundations could still falter.

How Leaders Can Capture Value

For business leaders, success requires focus and rigor. Relying on AI as a quick fix is a mistake.

Real change requires building a culture ready to adapt to AI, making focused bets on high-value domains, and embedding AI into operations. The companies that endured the dot-com collapse, such as [eBay](#) and [Netflix](#), did so by having a keen strategic focus and rapid operational adaptation. They resisted hype-driven diversions. Today, [Walmart](#) uses AI inventory bots to cut excess stock by 35% and boost accuracy by 15%, lowering costs while improving in-stock rates and customer experience. It also [just signed a deal](#) with OpenAI to allow users to buy its products directly through ChatGPT. The winners will embed AI where it reinforces their core advantage, not across every new trend.

Leaders should put resources into data governance and redesign organizations so people and AI can work together effectively. Personnel readiness involves continuous reskilling and evolving roles. Patience is vital since enterprise transformation is a multi-



year journey with value capture over time, instead of a quest for speculative gains.

If executives see AI only as a contest of spending, they miss the point. The challenge is to design governance that drives innovation and safeguards society. As I argued in [*Governance at a Crossroads*](#), AI policy works best when it evolves with the technology, pushing innovation forward while safety and fairness aren't lost. Regulatory sandboxes provide controlled environments for experimentation while maintaining accountability, and public-private partnerships can embed transparency and risk management.

Firms that engage early gain influence over how the AI economy is governed, helping ensure adoption is matched by resilience, fair oversight, and lasting trust in the technology.

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Artificial intelligence is both far-reaching and fragile. Its long-term trajectory points toward redefining industries, reshaping work, and altering the balance of global power. In the near term, geopolitical rivalry combines with deregulatory policy and speculative capital, creating conditions that strongly resemble past bubbles. The lesson of history is not that bubbles render



technologies worthless, but that they distort timing and expectations.

To ignore AI is to risk obsolescence. Success will come from disciplined adoption and pragmatic corporate governance. Leaders who focus on embedding AI into workflows and cultivating a workforce able to adapt will capture durable value. Those who engage early in emerging governance frameworks will help shape the rules of the AI economy. This moment will be remembered either as the start of an AI-driven economy or as overreach, depending on the decisions executives and investors make today.

More Resources

- [AI-Generated “Workslop” Is Destroying Productivity](#)
- [HBR’s Most-Read Gen AI Articles of 2025 So Far](#)
- [Addressing Gen AI’s Quality-Control Problem](#)
- [The Perils of Using AI to Replace Entry-Level Jobs](#)
- [What Companies with Successful AI Pilots Do Differently](#)
- [Don’t Let an AI Failure Harm Your Brand](#)





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